Dower Coppler: Signed Doppler Contrast for Intact-Skull Transcranial 3D Human fUS

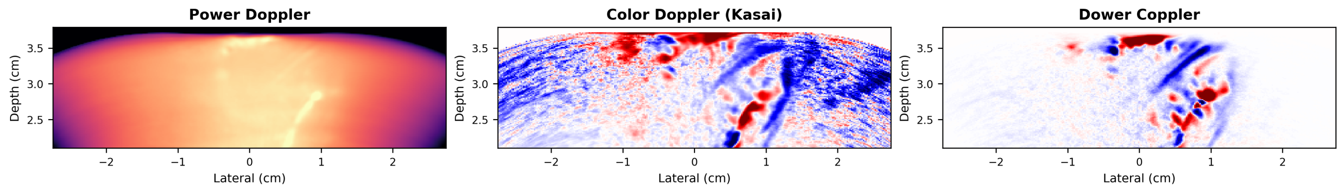

Figure 1: Intact-skull transcranial human cortex imaging from a 3D matrix-array acquisition (median of 480 Doppler buffers, plane-wave compounding). This panel shows a central reconstructed elevation plane from the lateral-elevation-depth Doppler volume. **Left**: power Doppler, displayed with a robust positive-pixel dB window, shows vessel structure but no directional information. **Center**: Kasai color Doppler ($\angle R_1$) reveals direction but is dominated by noise. **Right**: Dower Coppler ($v_\phi\, G_R\, R^2$, $K = 5$) combines multi-lag coherence weighting with directional information from the phase slope and fit quality. Opposing flow directions in adjacent cortical vessels are resolved in this intact-skull human transcranial acquisition.

Abstract

Ultrafast power Doppler imaging is sensitive to weak flow but discards flow direction, while classic autocorrelation color Doppler estimates signed velocity from phase and is sensitive to noise. These limitations are acute in intact-skull transcranial human 3D functional ultrasound (fUS): a matrix-array acquisition must preserve slow-time phase coherence after skull attenuation and aberration while reconstructing lateral, elevational, and depth-resolved Doppler volumes. We propose **Dower Coppler**, a composite signed Doppler index for SVD-filtered ultrafast compound data in this setting. Dower Coppler combines complex autocorrelation, multi-lag phase regression, power-like magnitude weighting, and fit-quality gating into a single scalar image. For each voxel, we estimate the beam-projected phase velocity v_ϕ from a weighted multi-lag phase fit, then multiply it by the geometric mean of the lag-correlation magnitudes and by the weighted R^2 of the phase fit, yielding $v_\phi\, \mathrm{gmean}_k(|R_k|)\, R^2$. In Monte Carlo simulations across five noise and interference scenarios, the multi-lag phase estimator reduces median absolute frequency error by 2–3× relative to the Kasai estimator. In a PyMUST-compatible parabolic-flow phantom with known velocity, the same multi-lag phase estimator reduces center-core velocity RMSE from $5.50\,\mathrm{mm\,s^{-1}}$ to $0.52\,\mathrm{mm\,s^{-1}}$ relative to lag-1 Kasai. We demonstrate the method *in vivo* on intact-skull transcranial human 3D fUS data, acquired without a cranial window. Across the reconstructed elevation volume, Dower Coppler reveals opposing flow directions in adjacent cortical vessels that are not visible in conventional power Doppler.

1 Introduction

Ultrafast ultrasound imaging, in which the entire field of view is insonified by unfocused or plane-wave transmissions at kilohertz frame rates, has enabled a new class of hemodynamic imaging modalities [1, 2]. With 2D matrix-array reception, the same ultrafast strategy can be extended from a single image plane to volumetric functional ultrasound (fUS), producing lateral, elevational, and depth-resolved Doppler volumes. Transcranial fUS exploits ultrafast sensitivity to image cerebral blood flow through the intact skull, but the transcranial setting imposes severe SNR constraints due to skull attenuation and aberration. The combination of intact-skull propagation and 3D matrix imaging is particularly demanding: each voxel must be reconstructed from an attenuated, aberrated aperture while preserving enough slow-time phase coherence for directional Doppler estimation. Spatiotemporal SVD clutter filtering is now a standard preprocessing step in ultrafast Doppler and fUS because it separates tissue and blood components using their different spatiotemporal coherence [1]. After clutter filtering, the most common flow-strength image is power Doppler, which displays integrated Doppler power rather than mean Doppler frequency [3]. Power Doppler is robust and straightforward to compute, but it discards phase information: the resulting image is unsigned and carries no information about flow velocity or direction.

The Kasai autocorrelator [4] recovers a signed velocity estimate from the phase of the lag-1 autocorrelation, $\angle R_1$, and complex autocorrelation methods form the classical foundation of color flow imaging. Extensions using arbitrary spatial and temporal autocorrelation lags have also been studied in the color flow literature [5]. Thus, neither SVD clutter filtering, autocorrelation velocity estimation, nor the use of multiple lags is new by itself. The practical problem we address is how to combine signed phase information with a power-like confidence image for low-SNR transcranial ultrafast data.

Shen and Guo [6] introduced temporal multiply-and-sum (TMAS), the direct product-of-lag-magnitudes precursor to our coherence term, showing that multiplicative combinations of slow-time autocorrelations can improve power Doppler signal-to-noise ratio. Motivated by that idea, we use multi-lag autocorrelations to separate coherent flow from incoherent background. Rather than multiplying all lag magnitudes directly, we use their geometric mean as a bounded coherence term. This keeps the lag-product intuition—pixels must be temporally coherent across multiple lags—without the rapidly growing dynamic range of a raw product. On its own, however, any magnitude-only lag statistic inherits the fundamental limitation of power Doppler: it discards phase and produces an unsigned map.

A natural remedy is to use the autocorrelation phases, not just the magnitudes. For a coherent scatterer moving at angular frequency ω, the lag-k autocorrelation satisfies $\angle R_k \approx \omega k$: the phase is linear in lag. Fitting a line to $\{\angle R_k\}_{k=1}^{K}$ yields a multi-lag frequency estimate that uses all available lags rather than just lag 1. The idea of estimating frequency from the slope of autocorrelation phase is classical—Tretter [7] analyzed its statistical properties, and Fitz [8] developed fast approximations. Dower Coppler should therefore be viewed as a composite Doppler index, not a new physical Doppler estimator from first principles.

In this paper we propose **Dower Coppler**: a multi-lag, coherence-weighted signed Doppler index for intact-skull transcranial 3D human fUS. It uses the signed axial phase velocity from a weighted multi-lag phase regression, multiplied by two confidence factors: the geometric mean of the lag-correlation magnitudes and the weighted R^2 of the phase-versus-lag fit. The regression weights are the autocorrelation magnitudes $|R_k|$, so that lags with stronger signal dominate the fitted phase slope, while pixels whose phase is not approximately linear in lag are attenuated by the fit-quality term. The distinctive contribution is this specific scalar product, $v_\phi \, G_R \, R^2$, applied to a transcranial human matrix-array Doppler volume rather than only to a single 2D imaging plane.

The experimental setting is also relevant. Human fUS studies have often used intraoperative imaging after skull opening [9] or imaging through acoustically transparent cranial implants/windows [10, 11]. More recently, contrast-free transcranial fUS through the intact adult skull has been demonstrated for cerebrovascular-reactivity measurements with power Doppler imaging [12]. Here, the data are also acquired transcranially through the intact adult skull, but with a matrix-array acquisition that preserves elevational information and with the goal of recovering signed flow-direction contrast in cortical vessels throughout a reconstructed 3D volume. This combination of intact-skull propagation, 3D matrix imaging, and directional Doppler estimation creates an extreme low-SNR operating point and motivates a conservative composite index that uses multi-lag evidence and explicit phase-fit quality rather than a single-lag color Doppler estimate alone.

We evaluate the index and its phase-velocity component in three settings: (1) Monte Carlo simulations under additive white Gaussian noise, decorrelation, lag outliers, clutter leakage, and mixed-flow scenarios; (2) a PyMUST-compatible synthetic slow-time flow phantom with a known velocity field; and (3) *in vivo* transcranial 3D functional ultrasound imaging of human cortex. Figure 1 (top of page) previews the key result.

2 Methods

2.1 Signal Model

Let $s_t(\boldsymbol{x})$ denote the clutter-filtered complex analytic signal at pixel $\boldsymbol{x}$ and slow-time frame t, where $t = 0, 1, \ldots, T-1$. For a coherent scatterer population moving at a constant Doppler angular frequency ω, the idealized signal is

$$s_t = A\, e^{j(\omega t + \phi_0)} + n_t, \tag{1}$$

where A is the (real, positive) amplitude, ϕ_0 is an arbitrary initial phase, and n_t is zero-mean additive noise.

The lag-k temporal autocorrelation is

$$R_k(\boldsymbol{x}) = \frac{1}{T-k} \sum_{t=0}^{T-k-1} s_{t+k}(\boldsymbol{x})\, s_t^*(\boldsymbol{x}). \tag{2}$$

In the noiseless, fully coherent case, $R_k \approx |A|^2 e^{j\omega k}$, so that

$$\angle R_k \approx \omega k. \tag{3}$$

2.2 Background: Multi-Lag Coherence

Standard power Doppler uses the zero-lag power $|R_0| = \frac{1}{T}\sum_t |s_t|^2$. A TMAS-style product of autocorrelation magnitudes across lags $k = 1, \ldots, K$ is

$$M_{\text{TMAS}} = \prod_{k=1}^{K} |R_k|. \tag{4}$$

This construction is motivated by the TMAS product of autocorrelation magnitudes [6]: because $|R_k|$ decays with lag for uncorrelated noise but remains roughly constant for coherent flow, the product suppresses background more strongly than $|R_0|$ alone. As motivated above, Dower Coppler instead uses the geometric mean of the same lag magnitudes,

$$G_R = \exp\left(\frac{1}{K}\sum_{k=1}^{K} \log\left(\max(|R_k|, \epsilon)\right)\right), \tag{5}$$

with $\epsilon = 10^{-30}$ for numerical stability, which preserves the multi-lag coherence weighting at a scale comparable to a single autocorrelation magnitude. Like the raw TMAS product, G_R is non-negative and carries no directional information.

2.3 Multi-Lag Phase Regression

We estimate the Doppler frequency from the slope of autocorrelation phase versus lag. Given the unwrapped phases

$$\phi_k = \text{unwrap}\big(\angle R_k\big), \quad k = 1, \ldots, K, \tag{6}$$

we seek the angular frequency $\hat{\omega}$ that minimizes a weighted residual. The regression is constrained through the origin ($\phi_0 = 0$ at lag 0), since R_0 is real and positive after clutter filtering.

Plain weighted least squares (WLS). The initial estimate uses the autocorrelation magnitudes as weights:

$$\hat{\omega}_{\text{WLS}} = \frac{\sum_{k=1}^{K} |R_k|\, k\, \phi_k}{\sum_{k=1}^{K} |R_k|\, k^2}. \tag{7}$$

This is the weighted regression of ϕ_k on k through the origin, where lags with stronger autocorrelation contribute more.

Fit quality. We quantify whether the phase sequence is well described by a single linear slope using a weighted coefficient of determination. With

$$\bar{\phi} = \frac{\sum_k |R_k|\, \phi_k}{\sum_k |R_k|}, \tag{8}$$

we compute

$$\text{SSE} = \sum_k |R_k|\, (\phi_k - \hat{\omega}_{\text{WLS}} k)^2, \tag{9}$$

$$\text{SST} = \sum_k |R_k|\, (\phi_k - \bar{\phi})^2, \tag{10}$$

and

$$Q = R^2 = \text{clamp}\left(1 - \frac{\text{SSE}}{\max(\text{SST}, 10^{-12})}, 0, 1\right). \tag{11}$$

Pixels with weak lag correlations or inconsistent phase evolution therefore receive low weight in the final image.

2.4 Dower Coppler

The Dower Coppler index combines the signed velocity estimated from the multi-lag phase slope with the multi-lag coherence and fit-quality terms. First, the phase slope in radians per slow-time frame is converted to a beam-projected axial phase velocity:

$$v_\phi = \hat{\omega}_{\text{WLS}}\, \frac{f_{\text{slow}}}{2\pi}\, \frac{c}{2 f_0}, \tag{12}$$

where f_{slow} is the cadence of the slow-time samples, c is the assumed speed of sound, and f_0 is the transmit center frequency. For compounded plane-wave data, f_{slow} is the transmit pulse repetition frequency divided by the number of transmissions per compounded frame. This is the standard axial Doppler conversion; no beam-flow angle correction is applied. The central composite index is

$$\boxed{I_{\text{Dower}}(p) = v_\phi(p)\; G_R(p)\; Q(p)}. \tag{13}$$

The result is positive for one axial flow direction and negative for the opposite direction, depending on the system sign convention. The sign comes from v_ϕ, while G_R and Q attenuate pixels with weak temporal coherence or poor phase linearity. Because G_R is signal-scale dependent, I_{Dower} should be interpreted as a signed Doppler evidence index rather than a quantitative velocity map; v_ϕ is retained separately when velocity is the quantity of interest.

Figure 2 isolates the contribution of the main terms in this composite index. The phase velocity v_ϕ alone contains the directional pattern but also substantial background phase noise. Multiplication by the multi-lag coherence strength G_R suppresses decorrelated background and emphasizes vessel-like structures. The R^2 panel shows the phase-linearity quality map, and multiplication by R^2 preserves the signed velocity interpretation while down-weighting pixels whose lag phases are poorly explained by a single slope. The G_R and R^2 panels are unsigned; direction enters through v_ϕ.

The per-pixel computational cost is small: the autocorrelations R_k are computed for K lags, followed by closed-form weighted least squares and a weighted R^2 calculation.

2.5 Phase Unwrapping

The autocorrelation phases $\angle R_k$ are individually wrapped to $(-\pi, \pi]$. Before regression, we apply a one-dimensional temporal unwrap along the lag axis: for each consecutive pair of lags, the phase difference is corrected to lie within $(-\pi, \pi]$ by adding integer multiples of 2π. This is a standard 1-D unwrap and does not require spatial information.

The final R^2 quality term attenuates isolated unwrapping failures: a miswrapped lag produces a large residual from the linear phase model, lowering the weighted fit quality for that pixel.

2.6 Use of AI Coding Assistants

AI coding assistants were used extensively across all tasks in this work, including the development of the processing and analysis code, the reproducibility scripts, and the drafting and editing of this manuscript. All methods, results, and claims were reviewed and verified by the authors, who take full responsibility for the content.

3 Simulation Study

3.1 Setup

We evaluate the phase slope estimators on synthetic slow-time ensembles of $T = 64$ frames. For each trial, a coherent signal is generated according to eq. (1) with unit amplitude and additive complex Gaussian noise at a specified SNR. The true angular frequency ω is drawn from the set $\{0.35, 0.8, 1.35\}$ rad/frame, and the SNR ranges from -12 to $+12$ dB. We compute lag autocorrelations up to $K = 5$ and run $N = 1024$ independent trials per condition.

Five scenarios are tested:

1. **AWGN**: coherent flow plus white Gaussian noise.
2. **Decorrelation**: flow with a slow phase random walk (models scatterer turnover).
3. **Lag outlier**: one high-lag autocorrelation is corrupted by a large additive phase error.
4. **Clutter leakage**: a residual slow-moving clutter component is added to the flow signal.
5. **Two-component flow**: a weaker flow component with the opposite sign is mixed with the primary flow.

3.2 Estimators Compared

- **Kasai**: $\hat{\omega} = \angle R_1$ (single-lag baseline).
- **WLS**: weighted least-squares phase slope (eq. (7)).
- **Huber-WLS**: robust comparison estimator using Huber residual weights ($\delta = 0.7$, 5 iterations).

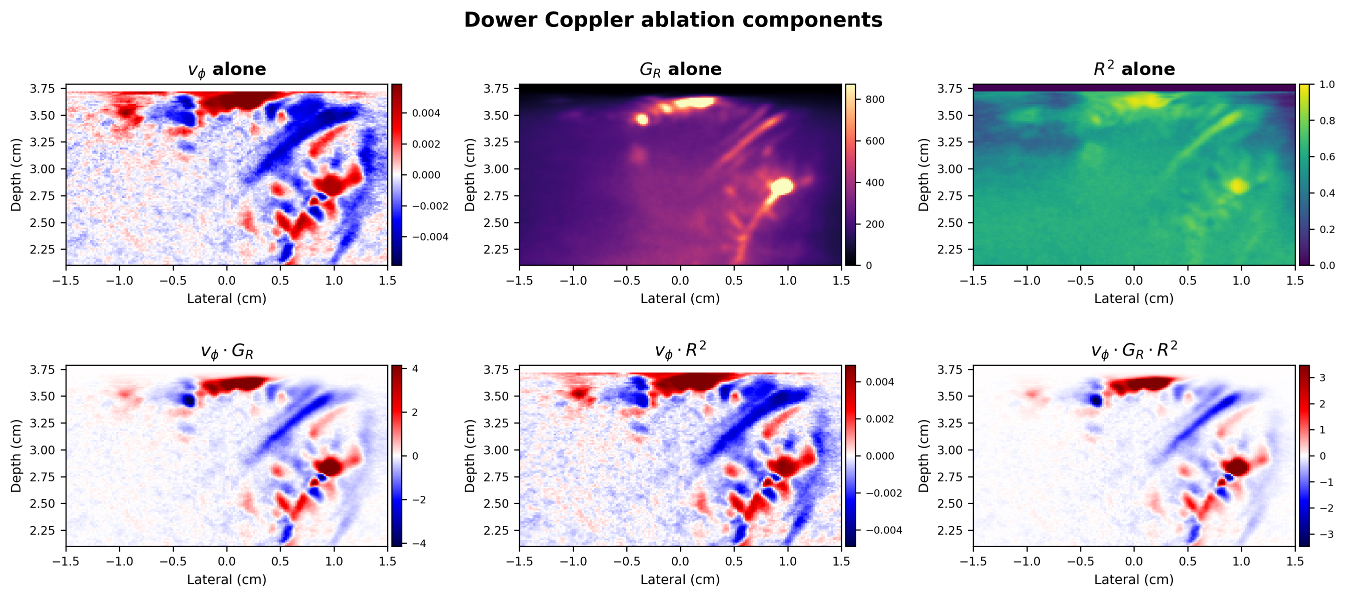

Figure 2: Ablation of the Dower Coppler components on the same transcranial image plane. Direction enters through v_ϕ; G_R and R^2 provide unsigned coherence and phase-linearity weights. The final panel uses $v_\phi G_R R^2$. Images are cropped to -1.5 to 1.5 cm laterally.

- **Circular-sign-WLS**: direction from a circular objective $\arg\max_\omega \sum_k |R_k| \cos(\phi_k - \omega k)$, magnitude from WLS.

3.3 Metrics

For each estimator, we report:

- **Median absolute error** (rad): median of $|\hat{\omega} - \omega|$.
- **RMSE** (rad): root mean squared error of the frequency estimate.
- **Sign error rate**: fraction of trials where $\mathrm{sgn}(\hat{\omega}) \neq \mathrm{sgn}(\omega)$.

3.4 PyMUST-Compatible Flow Phantom

As an additional spatially structured velocity check, we generated a PyMUST-compatible synthetic complex slow-time IQ phantom using the PyMUST L11-5v probe metadata and velocity scaling [13]. The phantom contained a single oblique parabolic flow target embedded in static tissue and weak incoherent background motion. We used $T = 96$ frames, a transmit center frequency of 7.6 MHz, an assumed sound speed of $1540\,\mathrm{m\,s^{-1}}$, a pulse-repetition frequency of 6000 Hz, and a peak axial velocity of $115\,\mathrm{mm\,s^{-1}}$. The lag-1 baseline was computed with PyMUST's `iq2doppler` implementation using a 3×3 Hamming neighborhood. The multi-lag estimate used the same filtered ensemble and a weighted phase fit over lags 1–5.

Because the synthetic velocity field is known, this phantom was evaluated by signed velocity error rather than by vessel/background CNR. We report error in two flow masks: the full target envelope, which includes parabolic low-velocity edge pixels, and a center-core ROI that excludes those edges. The center-core ROI is the cleaner test of quantitative velocity estimation; the full-envelope ROI is a stricter test of edge sensitivity.

3.5 Public-Benchmark External Check (PALA InSilicoFlow)

As a second, fully external velocity check we applied the same lag-1 Kasai and multi-lag phase-velocity estimators to the public PALA *InSilicoFlow* dataset (Zenodo record 4343435), a simulated microbubble flow sequence with ground-truth scatterer trajectories [14]. This dataset was not generated by us and provides an independent, third-party simulation. Its trajectory velocities, however, exceed the pulse-Doppler Nyquist limit of $12.32\,\mathrm{mm\,s^{-1}}$ at the dataset PRF and center frequency, so the underlying axial phase is aliased. We therefore compare each estimator against the trajectory-derived *aliased* axial phase velocity rather than the true velocity, and evaluate in two ROIs: raw localized positions with at least five samples, and a target-nearest mask. Estimator signs are multiplied by -1 to align the PALA increasing-z coordinate convention with the Doppler phase convention.

4 In Vivo Experiment

4.1 Ethics and Acoustic Safety

Human imaging was performed under WCG IRB Protocol #20254791. The volunteer provided informed consent prior to data acquisition. Ultrasound exposure was managed under the ALARA principle, using the lowest transmit level that produced usable Doppler data and avoiding unnecessary stationary insonation [15, 16].

Acoustic output was checked before the human acquisition using water-tank hydrophone measurements of the plane-wave transmit pressure. The largest measured peak pressure was bounded by 400 kPa. Expressed in the acoustic-output quantities used for diagnostic-ultrasound reporting, and using the measured pressure as a conservative upper bound on derated rarefactional pressure, this gives

$$\mathrm{MI} = \frac{p_{\mathrm{r},0.3}}{\sqrt{f_0}} \leq \frac{0.4}{\sqrt{2.75}} = 0.24, \tag{14}$$

where pressure is in MPa and f_0 is in MHz. Taking $Z = 1.5\,\mathrm{MPa\,s\,m^{-1}}$, the corresponding pulse-average intensity upper bound is

$$I_{\mathrm{SPPA}} = \frac{p^2}{2Z} \leq 5.4\,\mathrm{W\,cm^{-2}}. \tag{15}$$

With three transmit cycles at 2.75 MHz and a measured transmit pulse PRF of 1240.9 Hz, the duty factor is $3\,\mathrm{PRF}/f_0 = 0.00136$, yielding $I_{\mathrm{SPTA}} \leq 7.3\,\mathrm{mW\,cm^{-2}}$. These values are well below the FDA Track 3 global maximum acoustic-output levels for diagnostic ultrasound: derated $I_{\mathrm{SPTA}} \leq 720\,\mathrm{mW\,cm^{-2}}$ and either $\mathrm{MI} \leq 1.9$ or derated $I_{\mathrm{SPPA}} \leq 190\,\mathrm{W\,cm^{-2}}$ [16]. The measured-pressure estimates here satisfy both the MI and I_{SPPA} comparisons.

These calculations are conservative with respect to free-field pressure because no skull-transmission loss or diagnostic derating was credited. They are not, however, a substitute for a full IEC 62359 output-display characterization or a subject-specific thermal model of the skull [17, 18]. Because transcranial imaging places bone near the acoustic path, thermal risk was controlled operationally by the low duty factor, short acquisition windows, and ALARA-based dwell-time minimization.

4.2 Data Acquisition

Transcranial 3D functional ultrasound data were acquired from the temporal lobe of one healthy volunteer using a Butterfly iQ+ point-of-care matrix-array probe operated with custom firmware. The probe was placed over the temporal acoustic window and the acquisition was performed through the intact skull, without craniotomy, cranioplasty, or another acoustic cranial window. The matrix acquisition retained elevational aperture information, allowing the same ultratrace to be reconstructed as a lateral-elevation-depth Doppler volume rather than only a single 2D image plane. The source ultratrace contained 480 acquisitions. Each acquisition contained 700 compounded slow-time frames formed from five azimuthal plane-wave transmissions spanning $\pm 6°$; two additional noise loops were stored in the raw IQ data but were not used for Doppler processing. The transmit center frequency was 2.75 MHz with three transmit cycles, and the receive data were sampled at 4.17 MHz after decimation. The HDF5 metadata report an empirical pulse-repetition rate of 1240.9 Hz over the full ultratrace; because each compounded frame used five transmit angles, the corresponding compounded-frame cadence was approximately 248.2 Hz. The assumed sound speed was $1600\,\mathrm{m\,s^{-1}}$.

The original beamformed volume covered -27.456 mm to 27.456 mm laterally, -20 mm to 20 mm elevationally, and 21.0 mm to 37.896 mm in depth on a $147 \times 30 \times 58$ lateral-elevation-depth grid. The main paper figures were generated from a fine lateral/depth rebeamform of the two central elevation planes (original elevation indices 14 and 15), using all 480 acquisitions. This figure dataset used a $294 \times 2 \times 116$ lateral-elevation-depth grid over the same lateral and depth extent. The elevation-plane montage used a separate 10-plane fine-elevation rebeamform spanning $y = -4$ to 4 mm, also using all 480 acquisitions. The temporal-stability and split-half analyses used the available per-acquisition fine-elevation sidecar for acquisitions 200–400 and explicitly varied the number of buffers included in the median.

Table 1: Key acquisition and reconstruction parameters for the transcranial human fUS dataset.

Parameter	Value
Target / window	Temporal lobe / temporal acoustic window
Acquisitions in ultratrace	480
Frames per acquisition	700 compounded slow-time frames
Transmit angles	5 azimuthal angles over $\pm 6^\circ$
Transmit frequency	2.75 MHz, 3 cycles
Hydrophone pressure upper bound	400 kPa peak plane-wave pressure
Empirical pulse PRF	1240.9 Hz (full ultratrace)
Compounded-frame cadence	248.2 Hz
Sampling rate	4.17 MHz after decimation
Depth range	21.0 mm to 37.896 mm
Sound speed	$1600\,\mathrm{m\,s^{-1}}$
Original grid	$147 \times 30 \times 58$ (lat. $\times$ elev. $\times$ depth)
Figure grid	$294 \times 2 \times 116$ (lat. $\times$ elev. $\times$ depth)
Volume montage grid	$294 \times 10 \times 116$ (lat. $\times$ elev. $\times$ depth)

4.3 Processing Pipeline

Clutter filtering was performed with spatiotemporal SVD on a Casorati matrix: the complex compounded frames were reshaped with slow time as one dimension and image voxels as the other, low-rank tissue clutter components were removed, and the filtered data were reshaped back to the image grid. In the current processing, the compound data are mean-subtracted before SVD filtering; the lowest 8% of singular modes are removed, the high cutoff is 1.0, the fast SVD implementation is used, and the first five filtered frames are omitted from the power Doppler baseline summary. Beamforming used the MACH GPU delay-and-sum backend, channel mean subtraction, an f-number of 0.5, all enabled receive rows, and one-row compounding. The H5 acquisition metadata define a 30-plane elevation grid. The main fine-grid figure dataset rebeamformed the two central elevation planes at doubled lateral/depth sampling with 24 processing chunks and fast wavefront-fit downsampling of recovered full-2D transmit delays; the elevation montage used a separate 10-plane rebeamform spanning $y = -4$ to 4 mm. All Doppler estimators share the same filtered ensemble; they differ only in how the filtered slow-time data are reduced to an image. We computed:

- Standard power Doppler: the zero-lag filtered slow-time power, $P(p) = \sum_{t=6}^{T} |S'_t(p)|^2$.
- Kasai color Doppler: lag-1 velocity from $\angle R_1$ using the same axial Doppler conversion as in eq. (12).
- Multi-lag phase velocity from weighted phase regression.
- Dower Coppler, $v_\phi\, G_R\, R^2$, at $K = 5$.

5 Results

5.1 Simulation Results

Table 2 summarizes the estimator performance across all five scenarios.

Across all scenarios, WLS and Huber-WLS reduce the median absolute frequency error by approximately 2–3$\times$ relative to the Kasai estimator. In the clean AWGN case, WLS and Huber-WLS are nearly identical, confirming that the robust reweighting does not degrade performance when no outliers are present. Under the lag-outlier scenario, Huber-WLS achieves the lowest median error, while circular-sign-WLS gives the lowest RMSE and sign error in the displayed summary. The circular-sign-WLS hybrid consistently achieves the lowest sign error rate, particularly under clutter leakage and two-component flow.

As an additional spatial velocity check, the PyMUST-compatible parabolic flow phantom showed the same trend: in the coherent center-core ROI, the multi-lag phase estimator reduced signed velocity RMSE from $5.50\,\mathrm{mm\,s^{-1}}$ for lag-1 Kasai to $0.52\,\mathrm{mm\,s^{-1}}$, with near-zero bias ($0.10\,\mathrm{mm\,s^{-1}}$). This supports the use of the multi-lag slope as the signed velocity component of the composite index. Table 3 reports the full bias, MAE, and RMSE for both ROIs.

Table 2: Estimator performance averaged over SNR and velocity conditions. Best displayed values in each scenario are shown in **bold**; ties are determined after rounding to the shown precision.

Scenario	Estimator	Med. abs. err. (rad)	RMSE (rad)	Sign err. (%)
AWGN	Kasai	0.231	0.400	8.2
	WLS	**0.091**	0.258	6.6
	Huber-WLS	**0.091**	0.259	6.5
	Circ-WLS	0.095	**0.250**	**6.1**
Decorrelation	Kasai	0.246	0.422	8.3
	WLS	**0.110**	0.272	6.8
	Huber-WLS	**0.110**	0.272	6.7
	Circ-WLS	0.114	**0.267**	**6.5**
Lag outlier	Kasai	0.241	0.405	8.5
	WLS	0.108	0.280	7.2
	Huber-WLS	**0.105**	0.269	7.2
	Circ-WLS	0.107	**0.265**	**6.2**
Clutter leak	Kasai	0.385	0.553	10.2
	WLS	**0.180**	0.374	9.4
	Huber-WLS	0.181	0.374	9.2
	Circ-WLS	0.183	**0.370**	**8.6**
Two-component	Kasai	0.343	0.517	10.4
	WLS	**0.138**	0.325	9.4
	Huber-WLS	0.143	0.327	9.3
	Circ-WLS	0.140	**0.315**	**8.5**

On the fully external PALA InSilicoFlow benchmark (section 3.5), the two estimators were instead comparable rather than clearly separated (table 4). The multi-lag estimator gave a lower RMSE in both ROIs ($6.80\,\mathrm{mm\,s^{-1}}$ vs. $7.56\,\mathrm{mm\,s^{-1}}$ for the raw positions, and $7.28\,\mathrm{mm\,s^{-1}}$ vs. $8.40\,\mathrm{mm\,s^{-1}}$ for the target-nearest mask), but a slightly higher median absolute error and a lower linear correlation with the reference. We attribute the absence of a decisive separation to the aliased reference: because the simulated trajectory velocities exceed the pulse-Doppler Nyquist limit, the comparison is against an aliased axial phase velocity rather than the true velocity, so this benchmark is a weaker test of quantitative accuracy than the within-Nyquist PyMUST phantom. We include it because it is a public, third-party dataset and because reporting the comparable outcome alongside the favorable phantom result is the more complete summary.

5.2 In Vivo Results

On the displayed central elevation plane (fig. 1), power Doppler delineates the cortical vasculature with high sensitivity but no direction, while Kasai color Doppler reveals direction but is dominated by noise. The proposed Dower Coppler attenuates incoherent pixels using the multi-lag magnitude and fit-quality terms while recovering flow direction from the multi-lag phase slope, so that adjacent cortical vessels with opposing flow directions—invisible in power Doppler—are clearly separated.

Table 3: PyMUST-compatible signed velocity accuracy for the lag-1 Kasai estimator and the magnitude-weighted $K = 5$ multi-lag phase-velocity estimator v_ϕ. The synthetic slow-time phantom used a known parabolic axial-flow field with peak velocity $115\,\mathrm{mm\,s^{-1}}$. The full-envelope ROI includes low-velocity vessel edges; the center-core ROI excludes those edge pixels.

ROI	Estimator	Bias	MAE ($\mathrm{mm\,s^{-1}}$)	RMSE
Envelope	Kasai	-9.15	9.17	10.40
Envelope	Multi-lag v_ϕ	-5.26	5.50	8.27
Core	Kasai	-5.19	5.19	5.50
Core	Multi-lag v_ϕ	0.10	0.41	0.52

Figure 3 separates the signed velocity estimate from the Dower Coppler amplitude weighting. The multi-lag phase velocity preserves the dominant flow-direction pattern seen by the lag-1 Kasai estima-

Table 4: Zenodo PALA InSilicoFlow signed axial phase-velocity accuracy for lag-1 Kasai and the magnitude-weighted $K = 5$ multi-lag phase velocity v_ϕ. Trajectory-derived velocities exceed the $12.32\,\mathrm{mm\,s^{-1}}$ pulse-Doppler Nyquist limit, so errors are computed against aliased axial phase velocity. Estimator signs are multiplied by -1 to align the PALA increasing-z coordinate with the Doppler phase convention.

ROI	Estimator	Bias	MAE ($\mathrm{mm\,s^{-1}}$)	RMSE	r
Raw $n \geq 5$	Kasai	-0.66	4.14	7.56	0.451
Raw $n \geq 5$	Multi-lag v_ϕ	-0.38	4.64	6.80	0.310
Target-nearest	Kasai	0.72	4.94	8.40	0.257
Target-nearest	Multi-lag v_ϕ	0.58	4.89	7.28	0.163

tor, but the Bland–Altman comparison shows that the two velocity estimates are not interchangeable at high-magnitude pixels. In this dataset the mean bias is near zero, while the limits of agreement are set by pixels where the lag-1 phase and the multi-lag phase-slope fit disagree.

Figure 5 quantifies this comparison using contrast-to-noise ratio (CNR) and generalized CNR (gCNR) across six vessel ROIs exported from the Doppler CNR viewer. For each exported signal mask, we use the largest circular ROI whose center lies in the exported mask and whose pixels have at least 80% overlap with that mask. This avoids one-pixel ROIs for thin vessel masks while keeping the measurement localized to the selected vessel. The same vessel-free background rectangle is used for every signal ROI so that the CNR values differ only because of the selected vessel pixels and Doppler image. The vessel masks were drawn on a reconstruction of the central elevation planes from acquisitions 200–399 and then applied to the full all-acquisition figure dataset; this is valid only insofar as the vessel positions are stable across the ultratrace, which is supported by the temporal-stability and split-half analyses below.

Dower Coppler achieves median CNR 13.4 dB across the six tolerant circular vessel selections, with values ranging from 8.8–21.8 dB. Power Doppler has median CNR 12.4 dB (range -5.8–14.9 dB), and Kasai color Doppler has median CNR -8.7 dB (range -17.1–5.3 dB) when the signed Doppler maps are measured by magnitude, as in the viewer CNR calculation. Thus the corrected standard power-Doppler baseline is competitive for unsigned vessel detection, while Dower Coppler retains directional sign and suppresses incoherent signed background. The smallest tolerant circles contain five pixels, rather than the one-pixel ROIs produced by strict inscribed circles in thin segmented masks; we emphasize the median as the more representative summary statistic. The gCNR metric confirms that the selected vessels are well separated from the shared background region when measured by magnitude, with Dower Coppler at approximately 1.0 across all selections and power Doppler at median 1.0 despite one lower-contrast selection. The directional gCNR comparison uses the signed maps directly and therefore compares only Kasai color Doppler and Dower Coppler. The gCNR values are, however, near the saturated value of 1.0 for several selections, including the smallest five-pixel ROIs, where complete histogram separation from the background is close to trivial; we therefore treat gCNR$\approx$ 1.0 on small ROIs as weak evidence and rely on the CNR magnitudes and the split-half consistency analysis (Appendix A) as the more informative summaries.

We also checked temporal stability by recomputing Dower Coppler maps from nested acquisition windows in the available per-acquisition sidecar. Within that window, the vessel pattern and flow-direction assignments remained visually stable from 201 acquisitions down to as few as 10 acquisitions, with increased background noise in the shorter windows. Split-half repeatability and elevational-volume checks, together with this temporal-stability montage, are provided in Appendices A and B.

6 Discussion

Dower Coppler is best viewed as a composite Doppler index rather than a new Doppler estimator from first principles. Its ingredients—SVD clutter filtering, complex autocorrelation Doppler, multi-lag phase estimation, and power-like magnitude weighting—are well established. The distinct design choice is the scalar product $v_\phi\, G_R\, R^2$: signed phase velocity multiplied by multi-lag geometric coherence strength and by phase-linearity quality. This hybrid output is signed like color Doppler, amplitude-weighted like power Doppler, and confidence-gated by the quality of a single multi-lag phase-slope model.

Relationship to prior work. The individual ingredients are all established: SVD clutter filtering [1], Kasai autocorrelation velocity [4], multi-lag

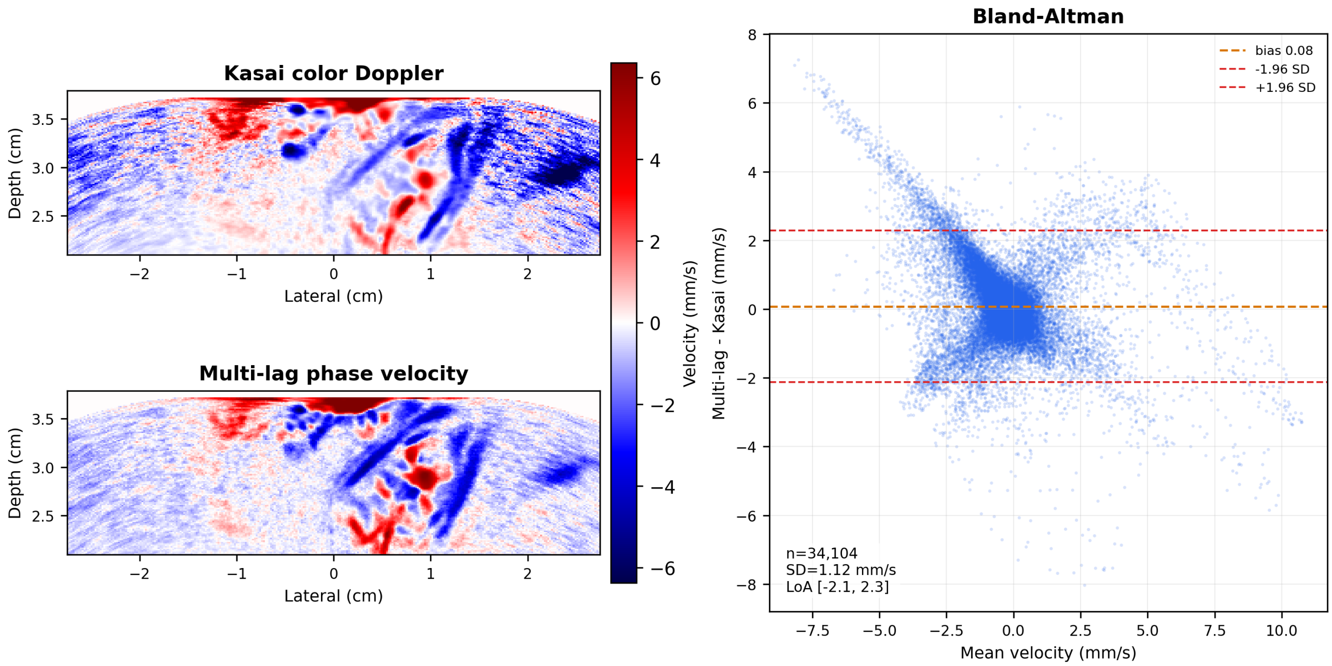

Figure 3: Comparison of conventional lag-1 Kasai color Doppler and the multi-lag phase-velocity estimate. The velocity maps use a common symmetric mm/s scale; the Bland–Altman plot compares all finite pixels in the displayed plane.

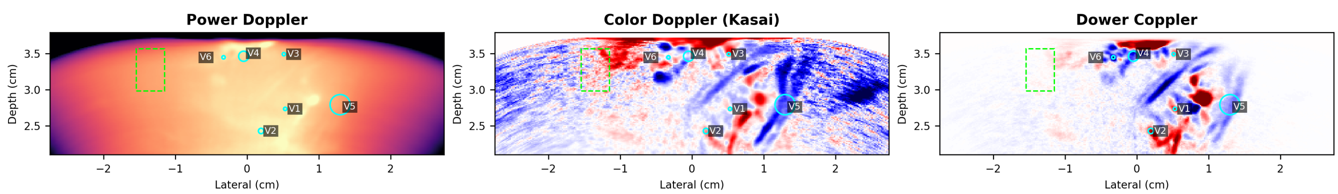

Figure 4: Regions used for the CNR and gCNR measurements in fig. 5. Cyan circles are tolerant circular signal ROIs centered inside the exported vessel masks with at least 80% mask overlap. The dashed green rectangle is the shared background ROI.

color-flow autocorrelation [5], phase-regression frequency estimation [7, 8], power Doppler [3], and the TMAS product-of-lag-magnitudes that motivates G_R [6]. Two close relatives are worth distinguishing: the Loupas two-dimensional autocorrelator uses temporal and depth lags jointly, but for axial frequency correction rather than multi-lag phase-slope fitting [19]; and multi-lag estimators in weather radar fit correlation models across lags rather than a Doppler phase line [20, 21]. Relative to all of these, the contribution is the specific coherence-weighted signed index, and in particular the per-pixel phase-linearity R^2 used as a confidence gate: unlike standard spectral-width or variance estimates, it measures how well the lag phases are explained by one linear slope, penalizing multi-component flow, turbulence, wrapping errors, and residual clutter. The claim is therefore not novelty for intact-skull adult fUS [12], autocorrelation Doppler, SVD filtering, or multi-lag correlation processing broadly.

Choice of parameters. The maximum lag $K = 5$ was chosen empirically: higher lags provide more fitting leverage but also more decorrelation, particularly through the skull. We used an SVD low cutoff of 0.08, high cutoff of 1.0, mean subtraction before SVD, and omitted the first five filtered frames from Doppler estimation. The current implementation uses weighted least squares rather than Huber reweighting for the final Dower Coppler image; Huber-derived quality maps remain useful diagnostics but are not part of the default formula.

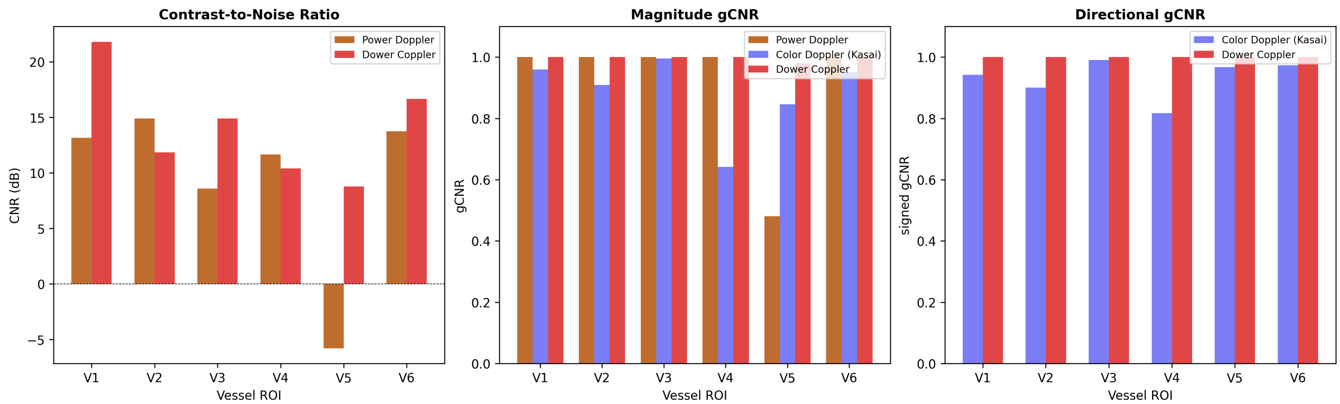

Figure 5: **Left**: CNR in dB across six vessel ROIs. Dower Coppler has median CNR 13.4 dB across the six selections (range 8.8–21.8 dB), compared with 12.4 dB for power Doppler. **Middle**: magnitude generalized CNR (1 − histogram overlap) using power Doppler, |color Doppler|, and |Dower Coppler|. **Right**: directional gCNR computed on the signed color Doppler and signed Dower Coppler values; power Doppler is omitted because it has no direction sign.

Limitations. This study uses data from a single volunteer with a single probe. The transcranial 3D matrix-imaging scenario—with low SNR, strong clutter, skull-induced aberrations, and elevation-dependent focusing errors—is a challenging test case, but validation on a larger cohort and with physical flow phantoms providing ground-truth velocity fields is needed. Dower Coppler is not itself a quantitative velocity image: G_R depends on gain, depth attenuation, beamforming normalization, SVD scaling, and backscatter strength. Quantitative velocity analysis should use v_ϕ directly, with appropriate attention to angle dependence and aliasing. The conversion in eq. (12) estimates the beam-projected axial component; it does not correct for beam-flow angle. Multi-lag fitting also does not remove the fundamental pulse-Doppler phase-wrap ambiguity: the lag phases still have to be unwrap-able. Finally, the method assumes a single dominant flow direction per pixel; bidirectional or turbulent flow can produce a compromised single-slope estimate, though the R^2 term attenuates pixels whose phase sequence is poorly described by a single velocity.

Extensions. The fit-quality metric Q can also be displayed as a standalone quality map, and robust alternatives such as Huber-weighted slopes or circular-sign hybrids remain useful variants where isolated lag outliers or sign accuracy dominate the design objective.

7 Conclusion

We have proposed Dower Coppler, a multi-lag coherence-weighted signed Doppler index for SVD-filtered ultrafast compound ultrasound, motivated by intact-skull transcranial 3D human fUS. The method estimates signed beam-projected phase velocity from a multi-lag autocorrelation phase fit and weights it by lag-correlation strength and phase-fit quality. It adds signed directional contrast at minimal computational cost while attenuating weak or inconsistent pixels. Simulations show that multi-lag phase regression reduces frequency estimation error by 2–3× relative to the single-lag Kasai estimator. In a structured velocity check, the multi-lag phase estimate reduces RMSE strongly in a PyMUST-compatible parabolic-flow phantom relative to lag-1 Kasai. The *in vivo* demonstration shows signed directional contrast from human cortex through the intact adult skull and across a reconstructed 3D matrix-array elevation volume. In this setting, the method resolves opposing flow directions in adjacent cortical vessels—information that is lost in conventional power Doppler.

Code, reproducible analysis scripts, and the derived NPZ/CSV inputs used for the manuscript figures and PyMUST velocity check are available in the standalone paper repository: `https://github.com/ennucore/dower-coppler`.

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Conflicts of Interest

The authors report no conflict of interest.

A Split-Half Consistency

As a repeatability check, we split the available September 21 per-acquisition sidecar into two non-overlapping windows (acquisitions 200–299 and 300–400), recomputed median Dower Coppler maps, and compared signs on the closest available elevation plane ($y = -0.44$ mm). Across the full plane, including low-amplitude background, the Dower Coppler sign agreed for 77.4% of 13,237 finite nonzero pixels. Restricting the comparison to the highest-magnitude pixels gave substantially higher agreement: 97.7% for the top 10% of pixels and 98.5% for the top 5%. In the tolerant circular vessel ROIs used for the CNR analysis, the sign agreed between halves for 98.0% of finite nonzero pixels (248 finite nonzero ROI pixels).

Figure A1 shows the corresponding split-half Dower Coppler images. The same vessel structures and dominant flow directions are recovered independently from the first and second halves of the ultratrace; disagreement is concentrated mainly in low-amplitude background and vessel-edge pixels.

B Elevation Checks

Figure A2 demonstrates the temporal stability of the Dower Coppler map by varying the number of acquisitions included in the median image. Within the available sidecar window, the vessel pattern and flow direction assignments are stable from 201 acquisitions down to as few as 10 acquisitions, with background noise increasing as fewer acquisitions are included.

Figure A3 shows the Dower Coppler map across elevation planes 2–7 from a finer elevational rebeamforming run spanning $y = -4$ to 4 mm. The directional vascular pattern is recovered across adjacent elevation planes rather than appearing only in one selected 2D slice. Figure A4 provides an additional qualitative 3D rendering of the reconstructed Doppler volume from two viewpoints. A separate-recording qualitative check is shown in fig. A5; it can be replaced through the figure-generation script without editing the manuscript.

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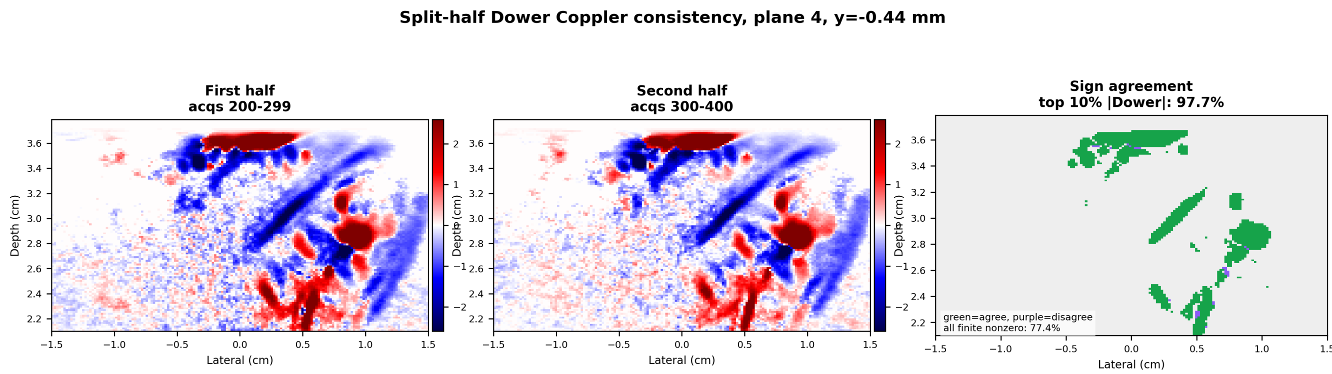

Figure A1: Split-half consistency of Dower Coppler on the fine-elevation plane at $y = -0.44$ mm. Left and middle panels show acquisitions 200–299 and 300–400 with a shared symmetric 97th-percentile scale. Right panel shows sign agreement within the top 10% of finite nonzero pixels by maximum absolute Dower magnitude.

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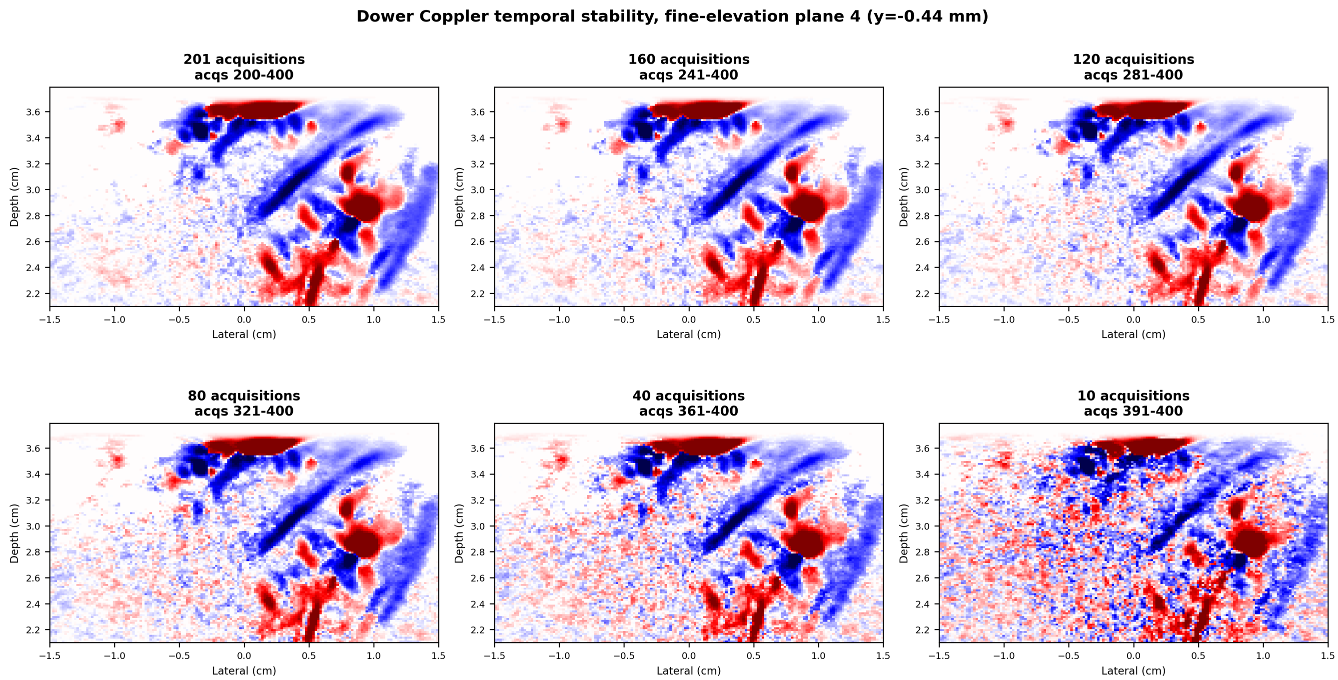

Figure A2: Dower Coppler maps computed from different numbers of acquisitions on the fine-elevation plane at $y = -0.44$ mm. All panels use one shared symmetric 97th-percentile color scale and are cropped to -1.5 to 1.5 cm laterally.

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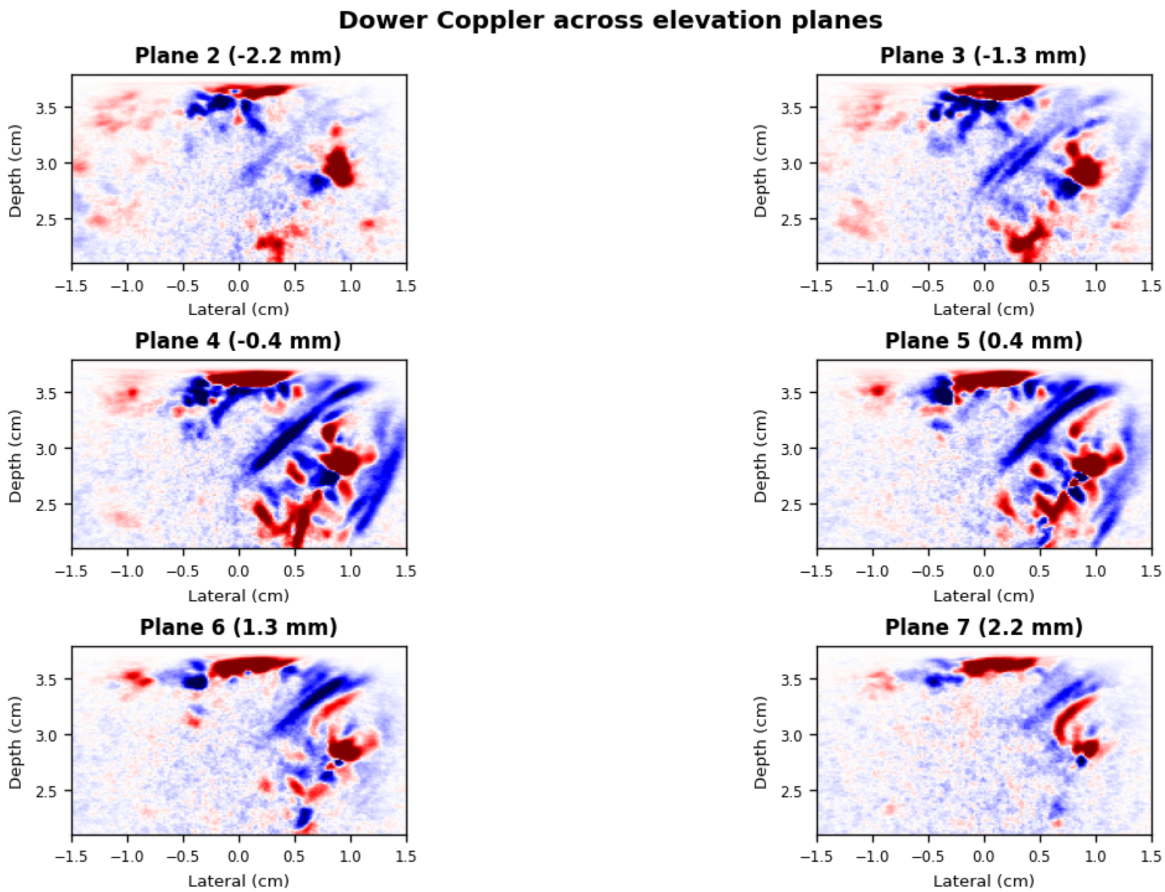

Figure A3: Dower Coppler maps across elevation planes 2–7 from the finer 3D matrix-array rebeamforming run spanning $y = -4$ to 4 mm. Panels use all 480 acquisitions and one shared symmetric 97th-percentile color scale. Images are cropped to -1.5 to 1.5 cm laterally.

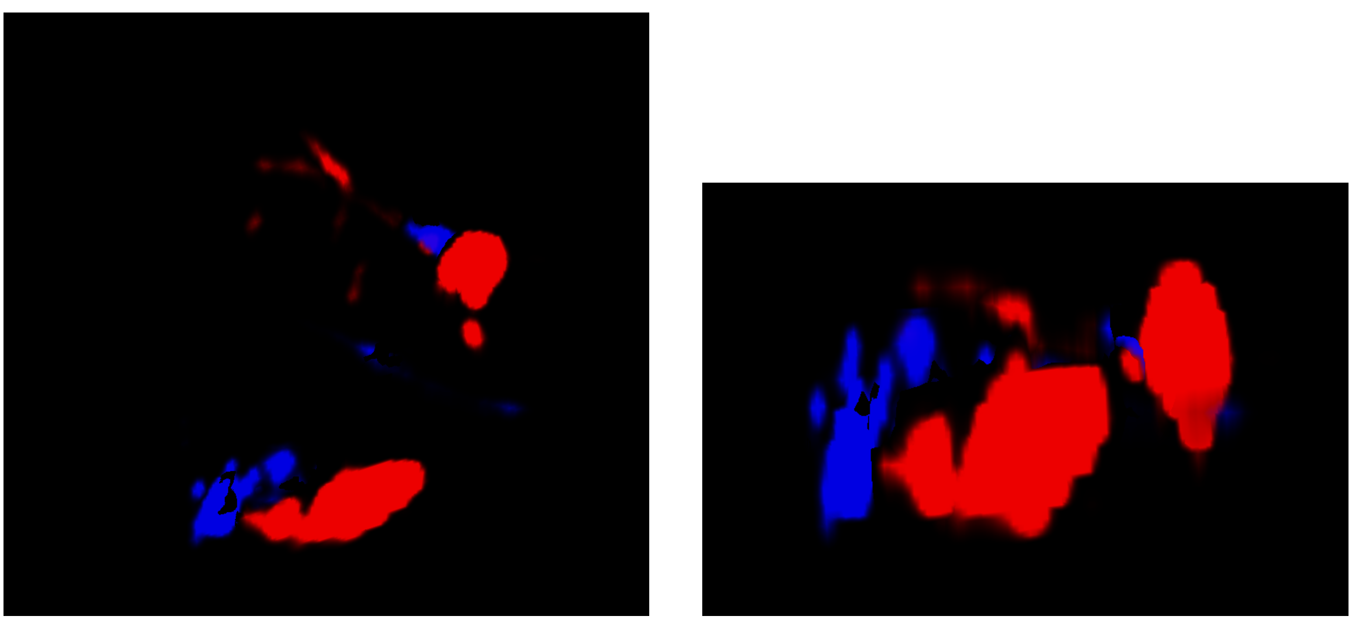

Figure A4: Qualitative 3D views of the reconstructed transcranial matrix-array Doppler volume. These renderings illustrate the lateral-elevation-depth structure underlying the elevation-plane maps in Figure A3.

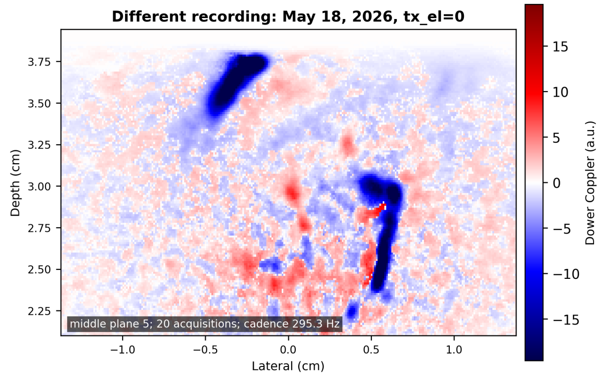

Figure A5: Qualitative check from a different May 18, 2026 recording using 20 acquisitions, compounded-frame cadence 295.3 Hz, $f_0 = 2.75$ MHz, $c = 1600$ m/s, SVD cutoff 0.08, and $K = 5$.